



AJAY PRASATH P 2024-CSE ▾

A2**Started on** Wednesday, 13 August 2025, 10:13 AM**State** Finished**Completed on** Wednesday, 13 August 2025, 11:27 AM**Time taken** 1 hour 13 mins**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
```

```
{
```

```
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
```

```
    {
```

```
        i++;
```

```
        s += i;
```

```
    }
```

```
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```

1  #include<stdio.h>
2  void function(int n){
3      int c=0;
4      int i=1;
5      c++;
6      int s=1;
7      c++;
8      while(s<=n){
9          c++;
10         i++;
11         c++;
12         s+=i;
13         c++;
14     }
15     printf("%d",c+1);
16 }
17 int main(){
18     int n;
19     scanf("%d",&n);
20     function(n);
21     return 0;
22 }
```

	Input	Expected	Got	
✓	9	12	12	✓
✓	4	9	9	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)